Food, Feeding and Digestion

The leech feeds by sucking the blood of cattle and other domestic animals. During feeding the leech attaches itself to its victim strongly by the posterior sucker. The leech makes a **triradiate** or **Y shaped incision** in the skin of the host by the jaws protruded through the mouth. The blood is sucked by muscular pharynx and the salivary secretion is poured.

The ingested blood is stored in crop chambers and its diverticulum. The blood passes from the crop into the stomach. Digestion takes place in stomach by the action of proteolytic enzyme. The digested blood is then absorbed slowly by the intestine. Undigested food is stored in rectum and egested through anus.

Leeches prevent blood clotting by secreting a protein called **hirudin**. They also inject an anaesthetic substance that prevents the host from feeling their bite.

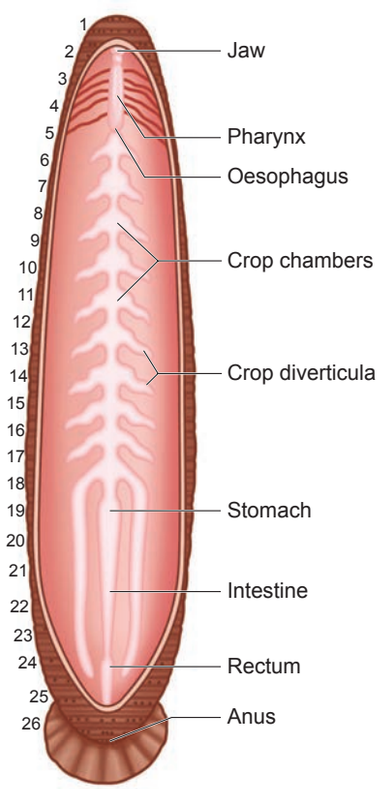


Figure 13.2 Digestive system of Leech

More to Know

- Leeches do not have ear, hence can sense vibrations through their skin.
- Leeches have 2 to 10 tiny eyes, which helps them to locate their food.
- Leeches can suck blood five times more than their body weight.
- It may take more than a year for the complete digestion and absorption of a full meal.

Table 13.1 Segmentation of Leech

External and Internal features	Segments in which the structures are present
Body segments	33
Anterior Sucker, Mouth, Eyes	1st - 5th
Posterior sucker	27th - 33rd
Pharynx	5th - 8th
Crop	9th - 18th
Stomach	19th
Intestine	10th - 22nd
Rectum	23rd - 26th
Anus	26th
Nephridiopores	6th - 22nd
Male genital aperture	10th
Female genital aperture	11th

13.1.7 Respiratory System

Respiration takes place through the **skin** in leech. Dense network of tiny blood vessels called as **capillaries** containing the haemocoelic fluid extend in between the cells of the epidermis. The exchange of respiratory gases takes place by diffusion. Oxygen dissolved in water diffuses through the skin into haemocoelic fluid, while

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